

Class-1

Assignment Questions:

1. Find the Fourier Transform of $f(t) = 5e^{2t}$, where $|t| < a$.

Ans: $F[f(t)] = \frac{10 \sinh(i\omega - 2)a}{i\omega - 2}$.

2. Find the Inverse Fourier Transform of the function $\frac{e^{-i\omega}}{2(1+i\omega)}$

Ans: $f(t) = \frac{1}{2}e^{-(t-1)}U(t-1)$.

3. Find the Fourier transform of $f(t) = 1 - t^2$, $|t| < 1$.

Ans: $F[f(t)] = 4 \left(\frac{\sin \omega - \omega \cos \omega}{\omega^3} \right)$.

4. Find $F^{-1} \left[\frac{1}{2 - 3i\omega - \omega^2} \right]$.

Ans: $f(t) = e^{-t}U(t) - e^{-2t}U(t)$.

5. Find $F^{-1} \left[\frac{1}{(9 + \omega^2)(4 + \omega^2)} \right]$ [Hint: $F^{-1} \left[\frac{2a}{a^2 + \omega^2} \right] = e^{-a|t|}$].

Ans: $f(t) = \frac{e^{-2|t|}}{20} - \frac{e^{-3|t|}}{30}$.

6. Find the Fourier Transform of $f(t) = a - |t|$, $|t| \leq a$.

Ans: $F[f(t)] = \frac{2(1 - \cos a\omega)}{\omega^2}$.

7. Find the Fourier Transform of $f(t) = e^{ikt}$, $a < t < b$.

Ans: $F[f(t)] = \frac{i}{k + \omega} [e^{i(k+\omega)b} - e^{i(k+\omega)a}]$